

The scalar product, magnitude and the angle between vectors

One multiplication that returns a number instead of a vector — and answers every angle question in space.

LESSON 7–8

2 × 40 MIN

GEOMETRY 11, §4

P1 · SCALAR PRODUCT

LESSON CLOCK

12'

20'

20'

18'

20'

10'

Two formulas for one product	12 min
The angle formula	20 min
Perpendicularity	20 min
Projections	18 min
Practice	20 min
Homework	10 min

LEARNING OBJECTIVES

- Compute the scalar product from components and from magnitudes and the angle.
- Find the angle between two vectors.
- Test two vectors for perpendicularity.
- Find the projection of one vector on another.

TEXTBOOK

National	pp. 35–46
Cambridge	Core & Extended, pp. 339–348
Workbook	Exercise 4.1

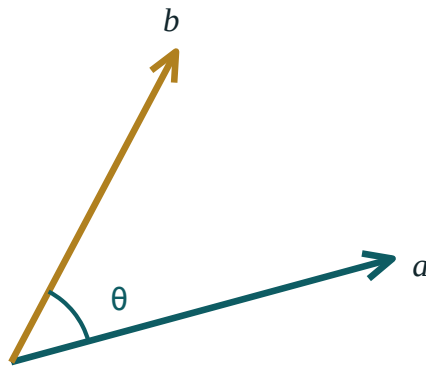
01

Explanation

Two formulas, one number

$$a \cdot b = a_1b_1 + a_2b_2 + a_3b_3 \text{ and } a \cdot b = |a||b| \cos \theta$$

The first computes it; the second interprets it. Setting them equal is what makes the angle findable.



$$a \cdot b = |a| |b| \cos \theta = a_1 b_1 + a_2 b_2 + a_3 b_3$$

perpendicular exactly when $a \cdot b = 0$

Two vectors and the angle between them, drawn from a common start point.

THE RESULT IS A NUMBER

$a \cdot b$ is a **scalar**, not a vector. Expressions like $(a \cdot b) \cdot c$ as a further dot product are meaningless — you cannot dot a number with a vector.

Properties: $a \cdot b = b \cdot a$, $a \cdot (b + c) = a \cdot b + a \cdot c$, and $a \cdot a = |a|^2$ — which gives a second way to find a magnitude.

The angle

$$\cos \theta = \frac{a \cdot b}{|a||b|}$$

For $a = (1, 2, 2)$ and $b = (2, -1, 2)$:

$$a \cdot b = 2 - 2 + 4 = 4, |a| = 3, |b| = 3 \Rightarrow \cos \theta = \frac{4}{9} \Rightarrow \theta \approx 63.6^\circ$$

$A \cdot B$	ANGLE	MEANING
> 0	$\theta < 90^\circ$	the vectors point broadly the same way
$= 0$	$\theta = 90^\circ$	perpendicular
< 0	$\theta > 90^\circ$	broadly opposite

THE SIGN ALONE TELLS YOU A LOT

Before computing an angle, look at the sign of $a \cdot b$. A negative dot product with a final answer of 40° is a certain arithmetic error.

Perpendicularity

THE SINGLE MOST USED CONSEQUENCE

$a \perp b \iff a \cdot b = 0$ (for non-zero vectors). No angles, no roots — one line of arithmetic.

$(1, 2, -1)$ and $(3, -1, 1)$: $3 - 2 - 1 = 0$, so they are perpendicular.

This is the test used for every right angle in the rest of the course — the diagonals of a rhombus, the edges of a cuboid, a line perpendicular to a plane.

Projection

The **projection** of a on b is the length of the shadow a casts along the direction of b :

$$proj = |a| \cos \theta = \frac{a \cdot b}{|b|}$$

For $a = (3, 4, 0)$ on $b = (1, 0, 0)$: $\frac{3}{1} = 3$ — the x -component, exactly as expected.

In physics the work done by a force F over a displacement s is $F \cdot s$ — the same idea, and the reason the scalar product was invented.

02 Worked examples**EXAMPLE 1** Find the angle between $a = (2, 1, -2)$ and $b = (1, 2, 2)$.

① $a \cdot b = 2 + 2 - 4 = 0$

② A zero dot product.

③ No further work is needed.

ANSWER 90° — they are perpendicular**EXAMPLE 2** Find the angle between $a = (1, 1, 0)$ and $b = (1, 0, 1)$.

① $a \cdot b = 1$

② $|a| = |b| = \sqrt{2}$

③ $\cos \theta = \frac{1}{2}$

④ $\theta = 60^\circ$

ANSWER 60°

EXAMPLE 3 Find t so that $(t, 2, 3)$ and $(4, t, -6)$ are perpendicular.

① $4t + 2t - 18 = 0$

② $6t = 18$

③ $t = 3$

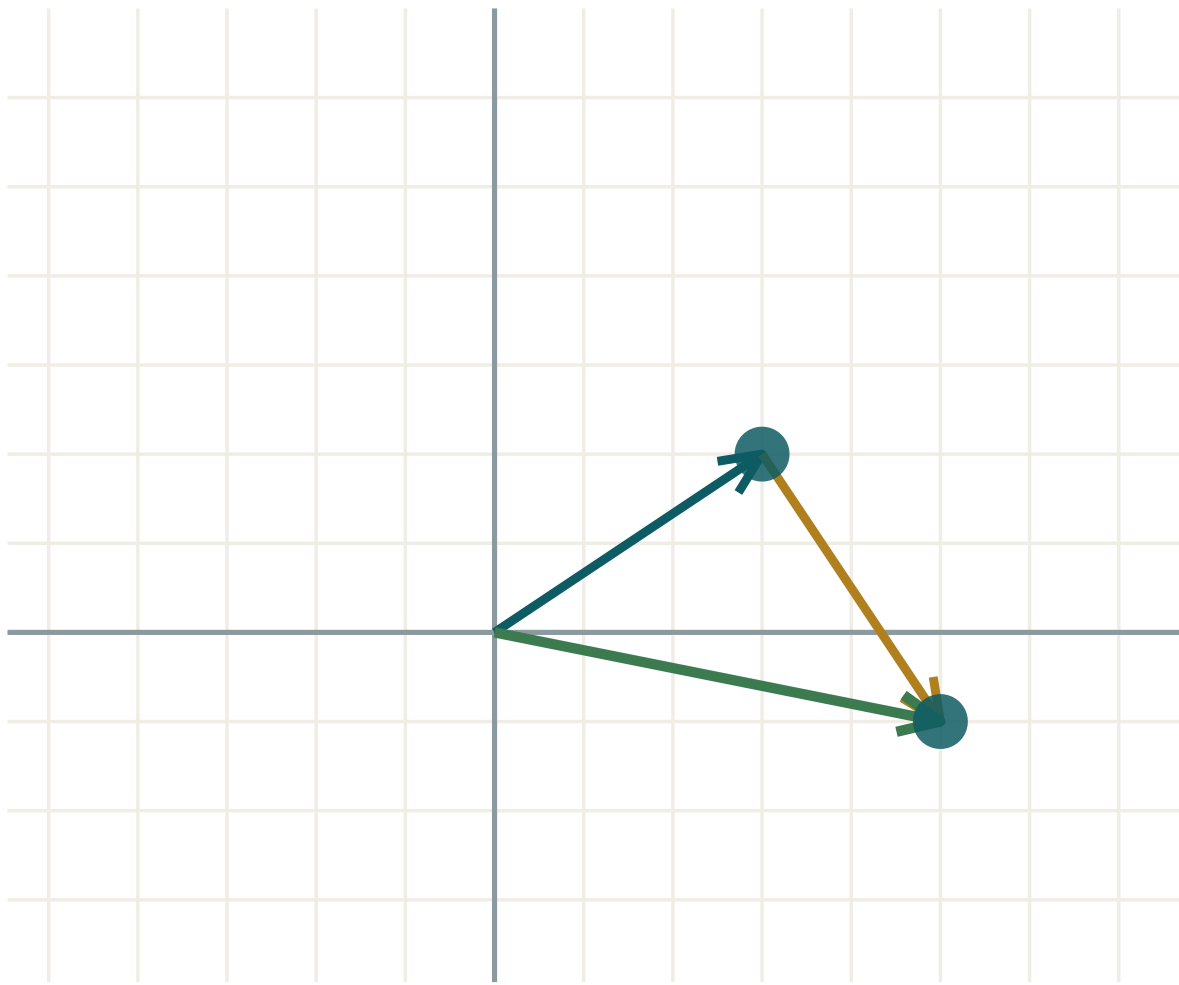
ANSWER

$t = 3$

03 Interactive model

Rotate one vector and watch the dot product pass through zero exactly at the right angle.

The angle between two vectors



$a \ (3; \ 2)$

$b \ (2; \ -3)$

$a + b \ (5; \ -1)$

$|a| \ 3.61$

$|b| \ 3.61$

$|a + b| \ 5.1$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Scalar (dot) product	Skalyar ko'paytma	Скалярное произведение
Angle between vectors	Vektorlar orasidagi burchak	Угол между векторами
Perpendicular vectors	Perpendikulyar vektorlar	Перпендикулярные векторы
Orthogonal	Ortogonal	Ортогональные
Projection	Proyeksiya	Проекция
Commutative	Kommutativ	Коммутативное
Distributive	Distributiv	Дистрибутивное
Direction cosines	Yo'naltiruvchi kosinuslar	Направляющие косинусы
Work done	Bajarilgan ish	Работа силы

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself** $a \cdot b$ is: $(1,2,3) \cdot (4,5,6)$ is: $a \cdot b = 0$ means:

$a \cdot a$ equals:

0

$|a|$

$|a|^2$

$2a$

If $a \cdot b < 0$ the angle is:

acute

right

obtuse

zero

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $(1,0,0) \cdot (0,1,0)$

ANSWER 0

2. $(1,2,3) \cdot (1,1,1)$

ANSWER 6

3. $(2,0,1) \cdot (3,4,5)$

ANSWER 11

4. $(1,1,1) \cdot (1,1,1)$

ANSWER 3

5. Are $(1,0,0)$ and $(0,0,1)$ perpendicular?

ANSWER yes

6. Angle between $(1,0,0)$ and $(1,0,0)$

ANSWER 0°

7. Angle between $(1,0,0)$ and $(-1,0,0)$

ANSWER 180°

Medium · the standard question

7 PROBLEMS

1. Angle between $(1,1,0)$ and $(1,0,1)$

ANSWER 60°

2. Angle between $(1,2,2)$ and $(2,-1,2)$

ANSWER $\approx 63.6^\circ$

3. Are $(2,1,-2)$ and $(1,2,2)$ perpendicular?

ANSWER yes

4. Find t : $(t,2,3) \perp (4,t,-6)$

ANSWER $t = 3$

5. Projection of $(3,4,0)$ on $(1,0,0)$

ANSWER 3

6. Projection of $(1,2,2)$ on $(0,0,1)$

ANSWER 2

7. $|a| = 5$, $|b| = 4$, $\theta = 60^\circ$; find $a \cdot b$

ANSWER 10

Hard · stretch and reason

7 PROBLEMS

1. Angle between the space diagonal and an edge of a unit cube

ANSWER $\approx 54.7^\circ$

2. Angle between two space diagonals of a unit cube

ANSWER $\approx 70.5^\circ$

3. Angle between a face diagonal and a space diagonal

ANSWER $\approx 35.26^\circ$

4. Find t so $(1, t, 2)$ makes 60° with $(1, 0, 0)$

ANSWER $t = \pm\sqrt{11}$

5. Prove $|a + b|^2 = |a|^2 + 2a \cdot b + |b|^2$

ANSWER Expand $(a+b) \cdot (a+b)$

6. Deduce the cosine rule from the scalar product

ANSWER $|a-b|^2 = |a|^2 + |b|^2 - 2|a||b|\cos \theta$

7. Prove the diagonals of a rhombus are perpendicular using vectors

ANSWER $(a+b) \cdot (a-b) = |a|^2 - |b|^2 = 0$

07 Homework

Homework — 5 tasks

Check the sign of $a \cdot b$ against your final angle every time.

- Find $a \cdot b$ and the angle for $a = (2, -1, 3)$, $b = (1, 4, 1)$.
- Find t so that $(3, t, -1)$ and $(2, 1, 5)$ are perpendicular.
- Find the angle between the space diagonal of a cube and one of its edges.
- Find the projection of $(4, 3, 0)$ on $(0, 1, 0)$.
- Prove that $a \cdot a = |a|^2$ and use it to find $|(2, 3, 6)|$.